

ANOVA (Analysis of Variance)

ANOVA is a statistical method to determine whether there is a significance differences between the means of three or more independence group by analyzing the variability within each group and between the groups.

It help in testing the null hypothesis that all group means are equal.

It does this by comparing two types of variation: (F-statistics)

1. Differences BETWEEN groups (how much group averages differ from each other)
2. Differences WITHIN groups (how much individuals in the **same** group vary naturally).

If the between-group differences are significantly larger than within-group variation, ANOVA tells us: At least one group is truly different. Otherwise, it concludes: The differences are likely due to random chance.

ANOVA Formula/ANOVA Table

Source of Variation	Sum of Squares	Degrees Of Freedom	Mean Squares	F-Value
Between Groups	$SSB = \sum n_1(\bar{x}_1 - \bar{x})^2$	$df_1 = k - 1$	$MSB = SSB / (k - 1)$	$f = MSB / MSE$
Error	$SSE = \sum \sum (\bar{x}_1 - \bar{x})^2$	$df_2 = N - 1$	$MSE = SSE / (N - k)$	
Total	$SST = SSE + SSB$	$df_3 = N - 1$		

There are mainly **two types** of [ANOVA](#):

1. [One-way ANOVA](#): Compares means of 3+ groups.

- **H₀**: All group means are equal
- **H₁**: At least one group differs

2. [Two-way ANOVA](#): Tests impact of two categorical variables and their interaction

- F = ANOVA Coefficient
- MSB = Mean of the total of squares between groupings
- MSW = Mean total of squares within groupings

- MSE = Mean [sum of squares](#) due to error
- SST = total Sum of squares
- p = Total number of populations
- n = The total number of samples in a population
- SSW = Sum of squares within the groups
- SSB = Sum of squares between the groups
- SSE = Sum of squares due to error
- s = [Standard deviation](#) of the samples
- N = Total number of observations

Assumption of ANOVA

1. ***Independence***: Observations are randomly sampled, and groups are independent.
2. ***Normality**: Residuals (errors) are approximately normally distributed (checked via Q-Q plots or Shapiro-Wilk test).
3. **Homoscedasticity***: Equal variances across groups (verified using Levene's or Bartlett's test).

Calculating ANOVA

Let's explore calculating ANOVA for the scenario:

Compare plant growth under 3 fertilizers (A, B, C):

- Fertilizer A: [10, 11, 12]
- Fertilizer B: [7, 8, 9]
- Fertilizer C: [4, 5, 6]

1. State Hypothesis

- Null Hypothesis (H_0): $\mu_A = \mu_B = \mu_C$
- Alternative Hypothesis (H_a): At least one μ differs.

2. Calculate Group means and Grand mean.

- Group Means: $\bar{X}_A, \bar{X}_B, \text{ and } \bar{X}_C$
- Grand Mean: $\bar{X}_{\text{grand}}$

$$\bar{X}_A = \frac{10+11+12}{3} = 11$$

$$\bar{X}_B = \frac{7+8+9}{3} = 8$$

$$\bar{X}_C = \frac{4+5+6}{3} = 5$$

$$\bar{X}_{\text{grand}} = \frac{10+11+12+7+8+9+4+5+6}{9} = \frac{72}{9} = 8$$

3. Compute Sum of Squares (SS):

SSB (Sum of Squares Between Groups): Accounts for variation due to the treatment or independent variable.

$$SSB = \sum n_i (\bar{X}_i - \bar{X}_{\text{grand}})^2$$

SSE (Sum of Squares Error or Within Groups): Accounts for variation within groups (random error or residuals).

$$SSE = \sum (x_i - \bar{X})^2$$

SST (Total Sum of Squares): Accounts for total variation from overall mean.

$$SST = SSB + SSW$$

$$SSB = 3(11 - 8)^2 + 3(8 - 8)^2 + 3(5 - 8)^2 = 3(9) + 3(0) + 3(9) = 54$$

SSE:

- Fertilizer A: $(10 - 11)^2 + (11 - 11)^2 + (12 - 11)^2 = 1 + 0 + 1 = 2$
- Fertilizer B: $(7 - 8)^2 + (8 - 8)^2 + (9 - 8)^2 = 1 + 0 + 1 = 2$
- Fertilizer C: $(4 - 5)^2 + (5 - 5)^2 + (6 - 5)^2 = 1 + 0 + 1 = 2$

$$SSW = 2 + 2 + 2 = 6$$

$$SST = 54 + 6 = 60$$

4. Calculate Degrees of Freedom (df):

df1 (Between Groups) = $k - 1$, where k is number of groups.

df2 (Within Groups) = $N - k$, where N is the total observations.

df3 (Total) = $N - 1$.

- $df1 = 3 - 1 = 2$
- $df2 = 9 - 3 = 6$
- $df3 = 9 - 1 = 8$

5. Calculate Mean Squares (MS):

MSB (Mean Square Between Groups) = $SSB / df1$.

MSE (Mean Square Error) = $SSE / df2$.

- $MSB = \frac{SSB}{df1} = \frac{54}{2} = 27$
- $MSW = \frac{SSW}{df2} = \frac{6}{6} = 1$

6. F-statistic:

The F-statistic is calculated as the ratio of **MSB** to **MSE**:

$$F = \frac{MSB}{MSE}$$

- $F = \frac{27}{1} = 27$

7. P-value:

The p-value is used to decide whether differences among groups are statistically significant. When the p-value is smaller than the significance level (α), the null hypothesis is rejected.

If $F > F_{critical} \rightarrow p < 0.05$: Null Hypothesis Rejected

Use the **F-distribution table** or software with: Numerator df1 = 2 , Denominator df2 = 6, $\alpha=0.05$

Critical F-value, $F_{critical}$: 5.14 (From F-distribution table)

$F > F_{critical}$: $27 > 5.14 \rightarrow p < 0.05$; Reject null hypothesis